



Wine Types & Quality

The Research Question

- ▶ Beyond just red and white wines, do wines form distinct groups based on their chemical makeup alone?
- ▶ Is there a relation between the quality of wine and its alcohol percentage?

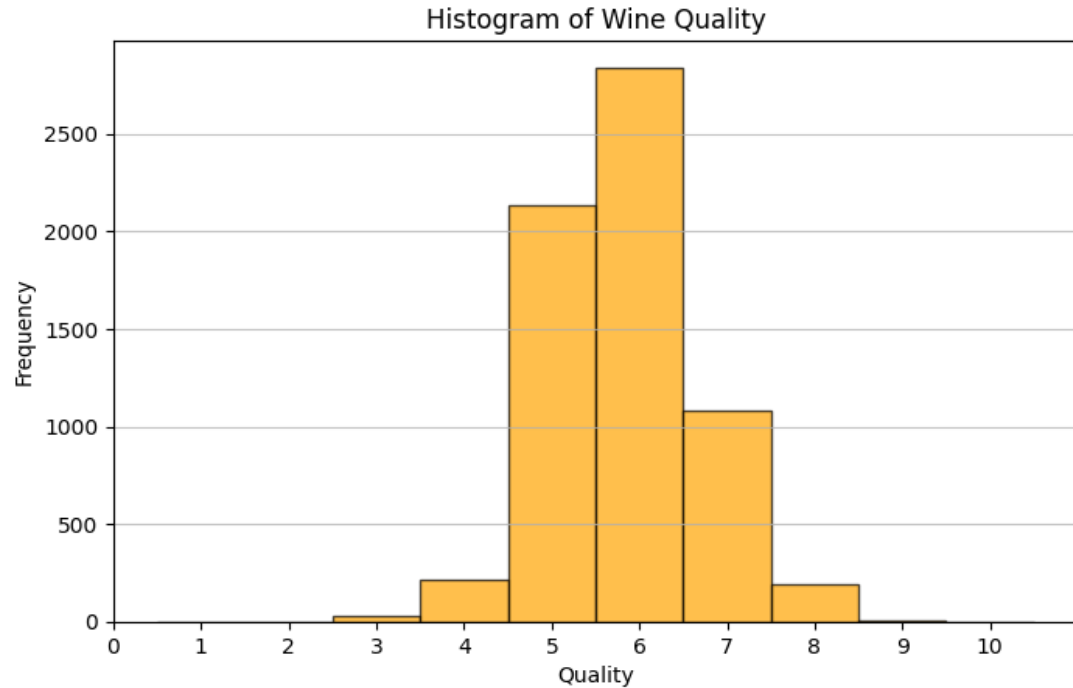
The Dataset

- ▶ UCI Repository - Wine Tasting
- ▶ 6497 samples
- ▶ 75%/25% white/red wine
- ▶ No missing values

Features
Fixed Acidity
Volatile Acidity
Citric Acid
Residual Sugar
Chlorides
<i>Free SO₂</i>
<i>Total SO₂</i>
Density
pH
Sulphates
Alcohol

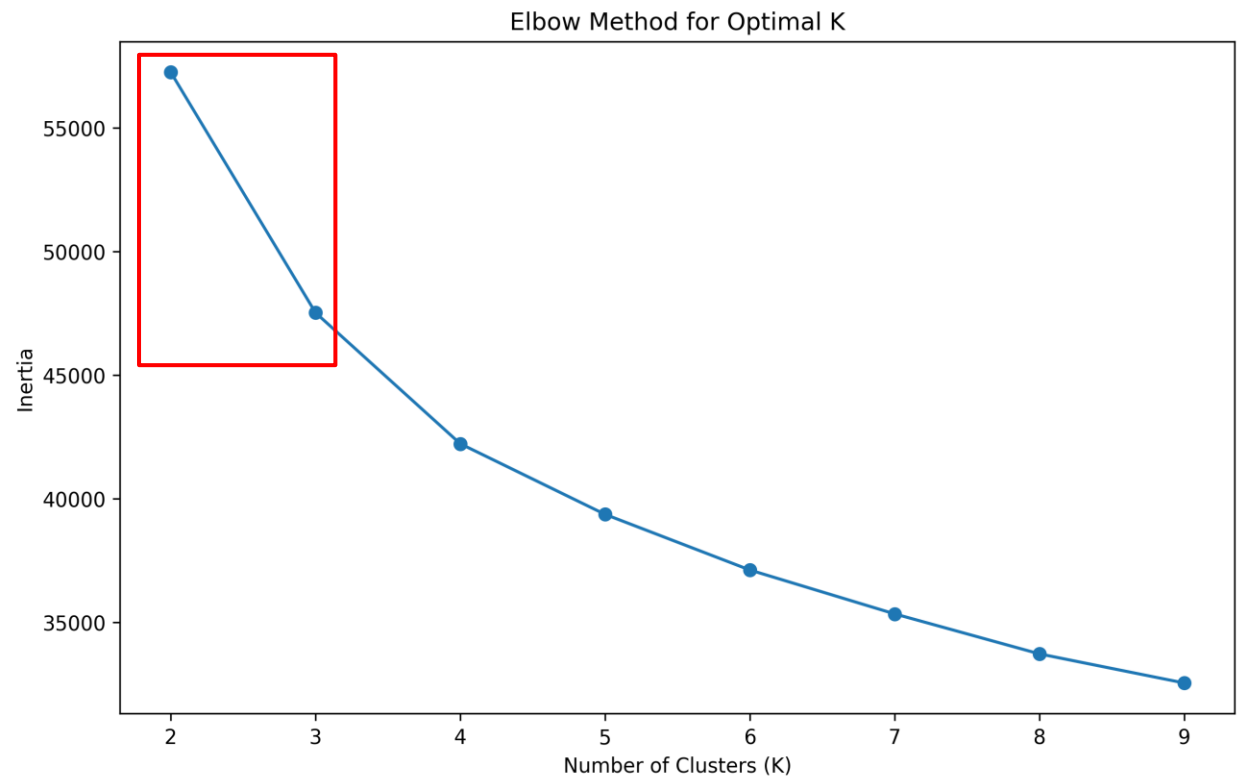
Further Exploring the Dataset

- The dataset only has samples with quality values between 3 and 9, and the labels are concentrated around the mean, approximating a normal distribution.



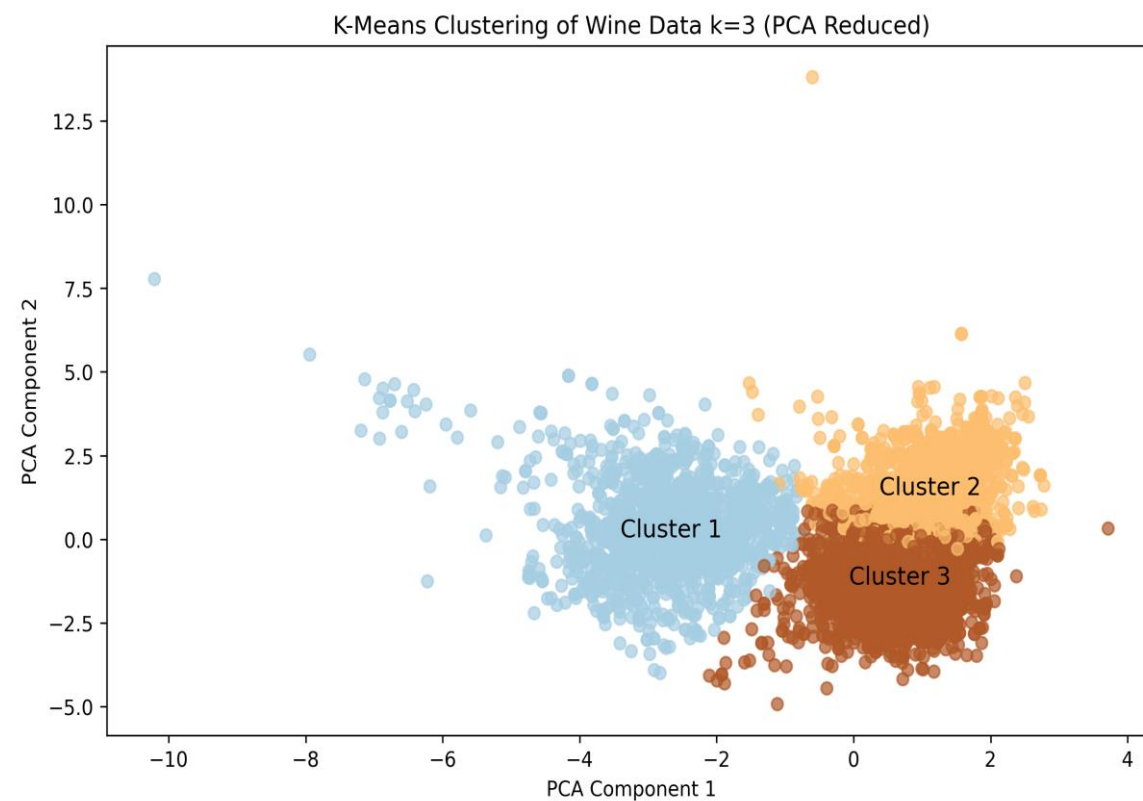
Clustering

- ▶ We chose the most suitable «natural» number of clusters by finding the greatest change in inertia (sum of distances between samples and their assigned cluster)
- ▶ Also, we standardize each feature before clustering, because all of them have different scales and units.



Interpreting the Clusters

Cluster	Mean Quality	White Wine %
1	5.61	1.59
2	5.61	99.58
3	6.05	98.44
Total	5.81	75.39



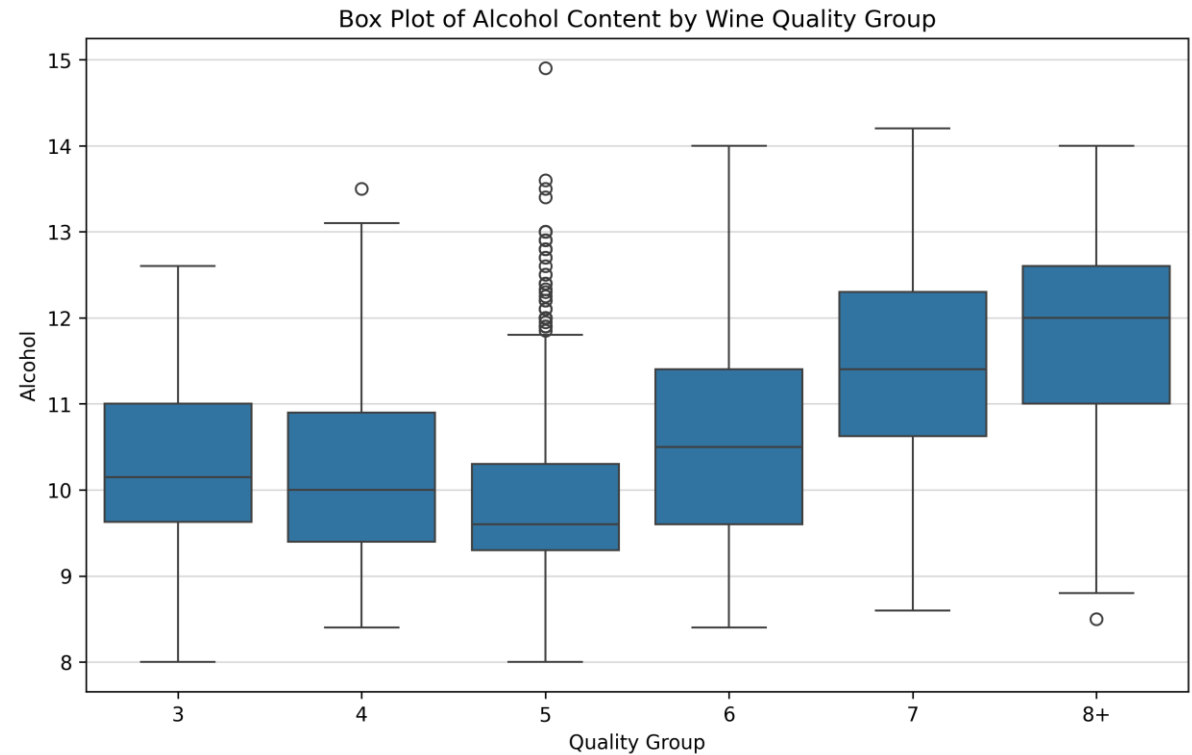
Interpreting The Clusters

- Cluster 1: Sour
 - (Highest in fixed acidity)
- Cluster 2: Sweet/Low Alcohol
 - (Highest in residual sugar)
- Cluster 3: Dry
 - (Highest in alcohol)

Cluster	Fixed Acidity	Volatile Acidity	Citric Acid	Residual Sugar	Chlorides	Free SO ₂	Total SO ₂	Density	pH	Sulphate	Alcohol
1	8.38	0.53	0.27	2.53	0.09	38%	46.29	1.0	3.31	0.66	10.36%
2	7.01	0.28	0.36	11.2	0.05	26%	167.29	1.0	3.15	0.49	9.53%
3	6.74	0.27	0.32	3.31	0.04	25%	119.22	0.99	3.21	0.49	11.17%

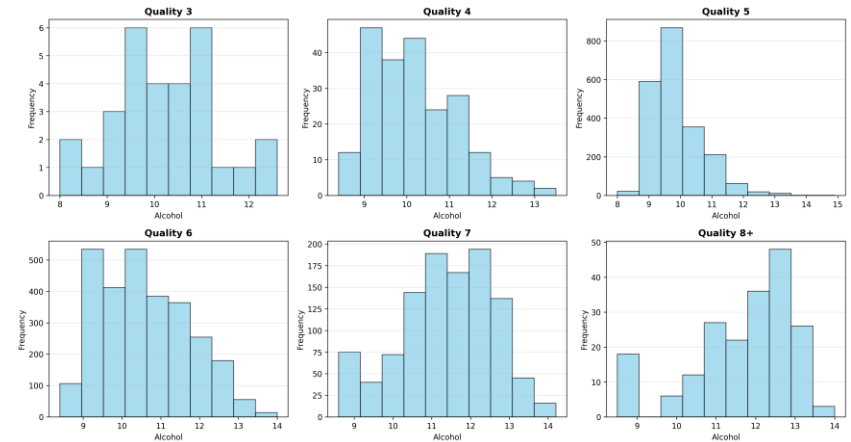
Alcohol Percentage vs Quality of Wine

► Data points with quality value above or equal to 8 are binned to the category 8+ as there were very few data points in these categories.



Checking Assumptions

- One-way ANOVA test makes a number of assumptions:
 - Normality: Even though the data for some of the quality groups are skewed, it's mostly normal
 - Constant variance: Lengths of the box plots are roughly the same.



Alcohol Percentage vs Quality of Wine

Anova Table Results

F-statistic	P-value
384.48	< 0.01

Tukey's HSD

Group 1	Group 2	Mean difference	P-value
3	7	1.171	< 0.01
3	8	1.4764	< 0.01
4	5	-0.3423	< 0.01
4	6	0.4075	< 0.01
4	7	1.2059	< 0.01
4	8	1.5113	< 0.01
5	6	0.7498	< 0.01
5	7	1.5482	< 0.01
5	8	1.8536	< 0.01
6	7	0.7985	< 0.01
6	8	1.1039	< 0.01
7	8	0.3054	< 0.01

Conclusions

- ▶ We could find clusters with noticeably different chemical properties in the dataset, forming different wine types.
- ▶ We could confirm our hypothesis that alcohol percentage is related to the quality value.



Thank you